Chapter 12 | Written solutions

Efficient Execution

Goal

Work through all eleven exercises, from a response timeline to an experimental protocol. The central idea is to retain enough information to combine score tiles without storing all their entries at once. Keep track of what a number measures: response time, aggregate throughput, stored bytes, or transferred bytes. Exercises 9-11 are optional. Both PDF editions contain every explanation and figure; the notebook edition also includes code and recorded outputs.

1. Draw a response timeline

Prefill consumes all four prompt tokens and builds their key–value history. The logits at the last prompt position select the first response token, y_1. It is available after 80 ms. The next call consumes y_1, appends its key and value, and selects y_2. Each later call consumes the previously selected ID.

Work just completed	Input consumed	Token selected	Elapsed time
Prefill	Four prompt tokens	y_1	80 ms
Decode 1	y_1	y_2	92 ms
Decode 2	y_2	y_3	104 ms
Decode 3	y_3	y_4	116 ms
Decode 4	y_4	y_5	128 ms

Output y_i is selected at $80 + (i - 1)12$ ms. Five outputs require only four calls after prefill. Feeding back the final y_5 would begin work on a sixth prediction. The **time to first token is 80 ms**, and the **total execution time is 128 ms**. Both start at the beginning of prefill. Sampling, queuing, and network time are outside this exercise.

2. Compare two batches

For one request, the run delivers five output tokens in 128 ms. For three batched requests, prefill selects three first tokens together at 100 ms. Each following 24 ms call advances all three requests by one token. The selection times for each request are 100, 124, 148, 172, and 196 ms.

Setup	Output tokens across run	Completion time per request	Aggregate output throughput
One request	5	128 ms	$5/0.128 = 39.0625$ tokens/s
Three requests in one batch	15	196 ms	$15/0.196 \approx 76.5306$ tokens/s

The numerator counts generated tokens only; the denominator includes prefill. The batch delivers three times as many tokens in $196/128 = 1.53125$ times the elapsed time. Its aggregate rate rises, but each user waits 68 ms longer. Dividing 196 ms by three does not give the wait experienced by a user: all three requests run concurrently and each finishes at 196 ms.

Per-request completion time answers how long one user waits for the whole response. Time to first token answers when text starts appearing: 80 ms versus 100 ms here. Throughput answers how much output the system delivers across its users per second. None of these invented numbers is a GPU benchmark.

3. Count one intermediate

One head has a score for each query–key pair, so its full score array contains $2048^2 = 4194304$ entries. Across 16 heads, that is 67,108,864 entries for one layer and one sequence. Multiply by bytes per entry:

Bytes per score	Bytes for one head	Bytes for 16 heads	MiB for 16 heads
2	8388608	134217728	128
4	16777216	268435456	256

A MiB is $2^{20} = 1048576$ bytes. The shape stays fixed, so doubling the entry size doubles the storage. A causal mask blocks some pairs but does not remove their slots from a materialized full square array.

This excludes the softmax probability array if it is also materialized, and the key and value arrays. It also excludes the query and output arrays, model weights, and other workspaces. Peak total memory depends on which arrays coexist, not just the size of one tensor.

4. Follow a changing maximum

Call scores $(0, \ln 3)$ with values $(2, 4)$ tile A, and score $\ln 6$ with value 1 tile B. After a nonempty set of keys, the state is

$$m = \max_j s_j, \qquad \ell = \sum_j e^{s_j - m}, \qquad u = \sum_j e^{s_j - m} v_j.$$

Both sums use the same reference m. If it rises to m', the identity $e^{s_j - m'} = e^{s_j - m} e^{m - m'}$ gives one multiplier for every old term. Rescale both old sums, then add the incoming terms at the new reference.

A, then B. Tile A sets $m = \ln 3$. Its shifted weights are $(1/3, 1)$, so $\ell = 4/3$ and $u = 2/3 + 4 = 14/3$. Tile B raises the maximum to $\ln 6$. The old sums receive a factor $e^{\ln 3 - \ln 6} = 1/2$, while B adds weight one and weighted value one:

$$\ell' = \tfrac{1}{2}\tfrac{4}{3} + 1 = \tfrac{5}{3}, \qquad u' = \tfrac{1}{2}\tfrac{14}{3} + 1 = \tfrac{10}{3}.$$

The final output is $(10/3)/(5/3) = 2$.

B, then A. Tile B starts at $(m, \ell, u) = (\ln 6, 1, 1)$. When A arrives, the maximum stays at $\ln 6$ and the old-state multiplier is one. A's new shifted weights are $(1/6, 1/2)$:

$$\ell' = 1 + \tfrac{1}{6} + \tfrac{1}{2} = \tfrac{5}{3}, \qquad u' = 1 + \tfrac{1}{6}(2) + \tfrac{1}{2}(4) = \tfrac{10}{3}.$$

Tile order	State after first tile (m, ℓ, u)	Multiplier on old sums at second tile	Final state
A, B	$(\ln 3, 4/3, 14/3)$	$1/2$	$(\ln 6, 5/3, 10/3)$
B, A	$(\ln 6, 1, 1)$	1	$(\ln 6, 5/3, 10/3)$

Direct softmax uses unshifted weights $(1, 3, 6)$, so the result is $(1 \cdot 2 + 3 \cdot 4 + 6 \cdot 1)/(1 + 3 + 6) = 20/10 = 2$. The online sums are both one-sixth of the unshifted sums; their ratio is unchanged. Different tile orders may change the last floating-point digits, but not the mathematical answer.

The independently normalized tile answers are $14/4 = 3.5$ for A and $6/6 = 1$ for B. Their ordinary average is **2.25**, which is wrong. A and B have exponential masses 4 and 6, respectively. The correct combination is $(4 \cdot 3.5 + 6 \cdot 1)/(4 + 6) = 2$. The relative masses are lost if we retain only each tile's normalized answer.

5. Check an uneven grid

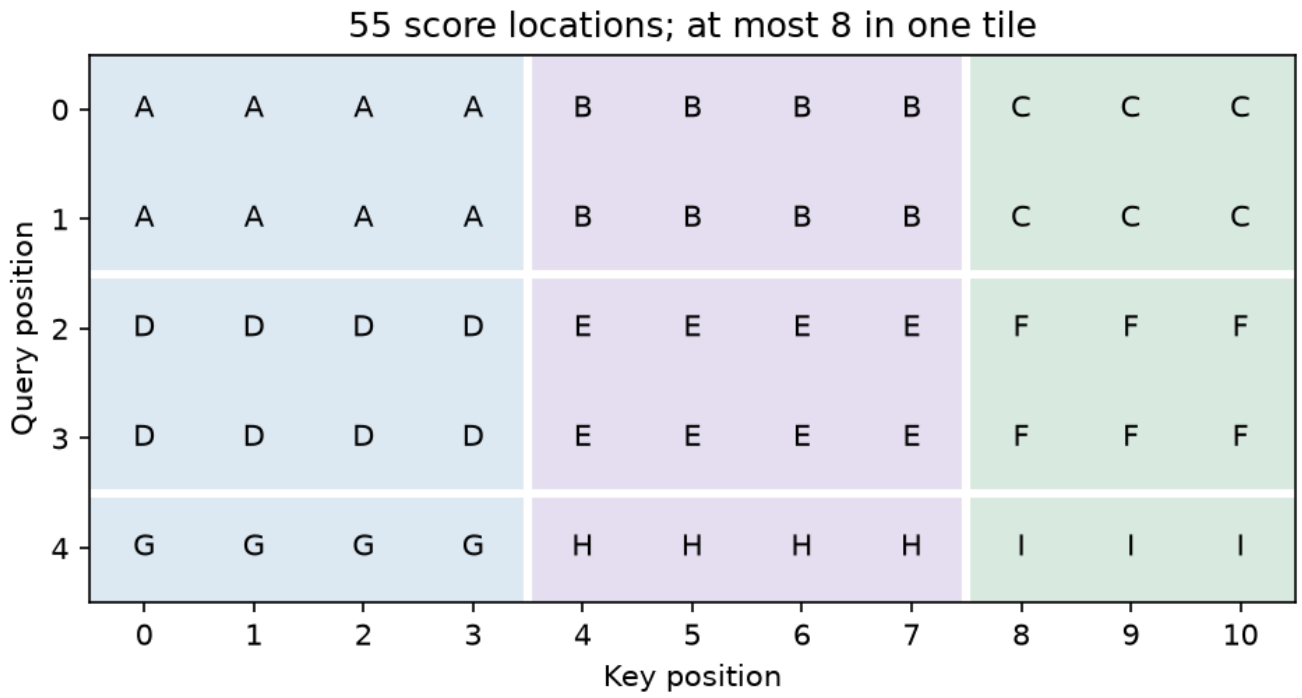
There are five query rows and eleven key columns. Each cell represents one query–key score. A tile contains up to two rows and four columns, so a tile that reaches the bottom or right edge may be smaller. We use zero-based row and column indices.

Query rows	Keys 0–3	Keys 4–7	Keys 8–10
0–1	A ů 2×4	B ů 2×4	C ů 2×3 (partial)
2–3	D ů 2×4	E ů 2×4	F ů 2×3 (partial)
4	G ů 1×4 (partial)	H ů 1×4 (partial)	I ů 1×3 (partial)

Each letter labels one tile. Tile C is partial because the final column group contains only three keys. Tile G is partial because the final row group contains just one query. Tile I has both short edges. The colored diagram below shows each score location and its tile.

Counting the whole grid gives $5 \times 11 = 55$ scores. Counting by tile also gives $4(8) + 2(6) + 2(4) + 3 = 55$, since every score belongs to exactly one tile. The largest tile holds $2 \times 4 = 8$ scores. Pass the lengths and then the two tile dimensions to `score_workspace` to check these two counts.

The sum counts every score location in the unmasked grid, even if the algorithm computes those scores at different times. It is also how many elements a single materialized score grid would store. The maximum counts the score elements in one tile at its largest. When tiles are processed sequentially, the algorithm can carry the running softmax state forward and release each tile's score array before moving on. It still visits all 55 scores in this example. Both counts are in elements. Multiplying by the score dtype's bytes per element gives the bytes for the corresponding score array alone. Inputs, outputs, masks, running accumulators, and any separate weight arrays also require space.



6. Change the tile shape

Three queries each have one feature, and the four keys and values each have one feature. Without a mask, every query reads all four keys. Each score is the query–key dot product divided by the square root of the key width. The width here is one, so for a scalar query q_i the scores are $q_i(0, 1, -1, 2)$.

Take the first query, $q_0 = 1$. Its scores are $(0, 1, -1, 2)$. The values at the same positions are $(4, 1, 3, 2)$, so the output is the sum of the four values multiplied by their normalized exponential weights.

$$y_0 = \frac{4 + e + 3e^{-1} + 2e^2}{1 + e + e^{-1} + e^2} \approx 1.9694644227.$$

We will compute a dense reference for all three rows. It subtracts the largest score in each row before exponentiating. This common shift leaves the softmax unchanged, since its factor cancels between the numerator and denominator.

During tiled attention, each query carries a running maximum m, an exponential sum ℓ, and a value-weighted sum u. After a set of scores has been seen, those sums contain e^{s-m} and $e^{s-m}v$ terms respectively. If a new tile raises m, both accumulated sums are rescaled to the new reference before its contributions are added. Every partition processes the same scores and values in this unmasked example. Once all keys have been seen, the ratio u/ℓ therefore gives the same result in exact arithmetic.

Compare each output with the dense reference and with the $(1, 1)$ result. We use float64, absolute tolerance 10^{-12}, and relative tolerance zero, since regrouping a sum can change the last digits in floating-point arithmetic.

All three partitions agree within the stated tolerance. They each cover a full grid of twelve scores. With $(1, 1)$, a tile holds just one score. With $(2, 3)$, the two query groups contain two and one rows, and the two key groups contain three and one columns. The four tiles contain $6, 2, 3, 1$ scores respectively, so the largest has six. The $(3, 4)$ tile covers the whole grid and holds all twelve scores at once. These are counts for the score array alone. Tile dimensions can change the size and number of intermediate arrays, so the largest score tile alone cannot tell us the peak memory or which shape runs fastest.

The common output vector is approximately $(1.9694644227, 1.9167694979, 1.9575193731)$. The largest score tiles contain 1, 6, and 12 entries, respectively; the full grid always contains 12.

7. Pass through an empty tile

Imagine a query whose first score tile contains two future keys. The dot products in that tile are 4 and 7, and the corresponding values are 8 and 5. The Boolean mask `[[False, False]]` excludes both keys. Neither their scores nor their values should enter the softmax calculation.

For one query, the running state contains a maximum score m, a denominator ℓ, and a weighted numerator u. The denominator adds up the exponential weights of permitted keys. The numerator adds the same weights times the corresponding values. The two sums use scores measured relative to m. The initial state has $m = -\infty$ and $\ell = u = 0$ because it has seen no permitted keys. The state arrays have shapes `[1]`, `[1]`, and `[1, 1]`, respectively. The numerator has an extra axis for its one value feature.

Here the first tile leaves that state unchanged. Internally the helper substitutes $-\infty$ for each masked score. For an empty row it uses zero as the temporary subtraction reference, so exponentiating a masked score gives $\exp(-\infty - 0) = 0$. This avoids subtracting $-\infty$ from itself and still records $-\infty$ as the stored maximum.

The next tile contains one permitted key, with score 2 and value 9. The first finite maximum is $m = 2$. The new key has weight $\exp(2 - 2) = 1$, giving $\ell = 1$ and $u = 9$. The old state contributes zero because it contains no earlier permitted key. The output becomes defined when the first valid tile makes ℓ positive. In a full tiled read, divide only after all tiles have contributed, since later valid keys can change the final ratio.

8. Trace a rectangular causal mask

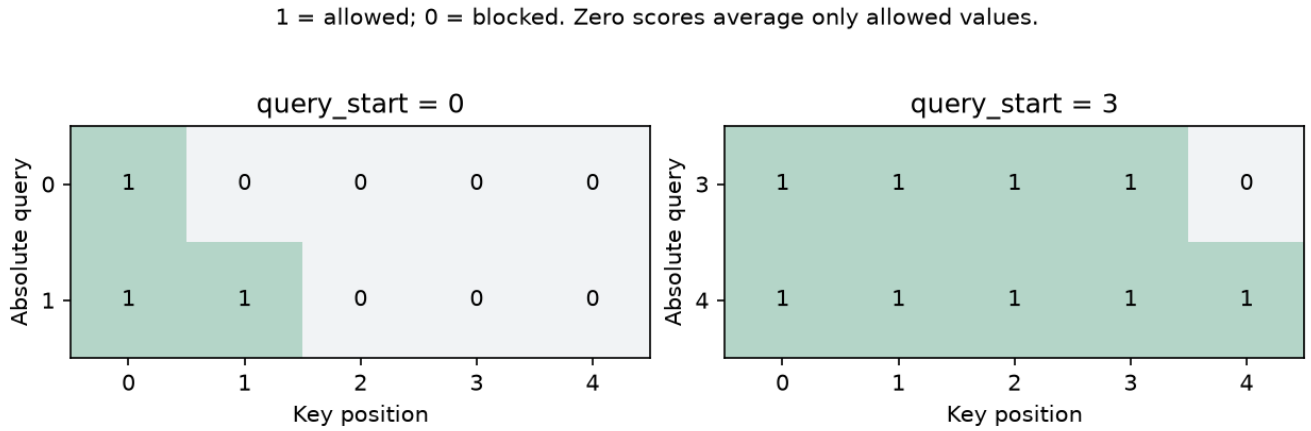
There are two query rows and five key rows. The query row within the current chunk has index $i \in \{0, 1\}$. Its position in the whole sequence is $p = \text{query_start} + i$. The key history starts at zero, so a key with index j is visible exactly when $j \le p$. In particular, the query can read the key at its own position.

All query and key vectors are zero. Every permitted score therefore equals zero, and every permitted key gets the same weight. With r permitted keys, each has softmax weight $\exp(0)/(r \exp(0)) = 1/r$. The output is the mean of the visible values, which makes the effect of the mask easy to check by hand.

Query start	Query row	Absolute position	Readable keys	Output
0	0	0	$[0]$	1
0	1	1	$[0, 1]$	$(1 + 3)/2 = 2$
3	0	3	$[0, 1, 2, 3]$	$(1 + 3 + 7 + 2)/4 = 3.25$
3	1	4	$[0, 1, 2, 3, 4]$	$(1 + 3 + 7 + 2 + 9)/5 = 4.4$

With one query row and two key columns per tile, the key slices are `[0, 1]`, `[2, 3]`, and `[4]`. For absolute position 0, the first tile admits only key 0 and the later tiles admit no keys. For absolute position 3, the first two tiles are fully visible, and the last tile is still in the future. For a chunk starting at position 3, the causal boundary shifts to key columns 3 and 4.

The two calls pass the same vectors to the same implementation. Only the absolute position of each query changes. The outputs at start 3 incorporate more of the history because those rows are later in the sequence. If a chunk actually belongs at positions 3 and 4, passing a start of zero wrongly removes three keys from each query. The float64 comparisons above allow an absolute difference of 10^{-12} and use zero relative tolerance.



9. Stress the exponentials (optional)

Give the first score tile $[800, 799]$ and the second tile $[801]$. The corresponding values are $(1, 3)$ and (5). The exponential of a score around 800 exceeds the largest representable float64 number. Online softmax avoids that calculation by subtracting the running maximum before exponentiating.

Write $a = e^{-1}$ and $b = e^{-2}$. The first tile sets $m = 800$. The two shifted scores are 0 and -1, so $\ell = 1 + a$ and $u = 1 + 3a$. When the second tile raises the maximum to 801, multiply both old sums by $\exp(800 - 801) = a$. The new score equals the new maximum and adds weight 1 and weighted value 5. The final state is

$$m = 801, \qquad \ell = a(1 + a) + 1 = 1 + a + b, \qquad u = a(1 + 3a) + 5 = 5 + a + 3b.$$

The direct stable calculation subtracts the final maximum from all three scores at once. It exponentiates the shifted row $(-1, -2, 0)$ and gives the same ratio $(5 + a + 3b)/(1 + a + b)$.

Adding 50 to all the scores also adds 50 to every finite running maximum. The shifted scores within each tile and the difference between the old and new maxima stay the same. Predict that both finite maxima increase by 50 and that both pairs of sums, as well as the final output, agree between the two runs.

The stored maxima move from $(800, 801)$ to $(850, 851)$. The denominator and numerator after each tile agree across the two runs, and the final states are finite. Both outputs match the directly stabilized calculation within an absolute tolerance of 10^{-12} with zero relative tolerance in float64. The same comparison works without forming an exponential of any of the original large scores.

The output for both score shifts is approximately 3.841024969440. Subtracting the maximum prevents overflow; shifting all scores by 50 changes the stored reference but not the relative weights.

10. Explain a changing bottleneck (optional)

With n prompt positions, prefill has n queries and n keys per head: a score grid of shape $n \times n$. The causal mask permits the diagonal and everything below it, for $n(n+1)/2$ pairs. A full allocation still has n^2 slots, including blocked entries.

The first cached call consumes the token selected after prefill. This token has absolute position n. Its layer computes one query and appends one key and value to the n stored positions. The query reads keys 0 through n, including its own key. Its score grid is $1 \times (n+1)$. For a four-position prompt, write 1 for an allowed read and 0 for a blocked read:

Stage / query	Key 0	Key 1	Key 2	Key 3	Key 4
Prefill, query 0	1	0	0	0	Not present
Prefill, query 1	1	1	0	0	Not present
Prefill, query 2	1	1	1	0	Not present
Prefill, query 3	1	1	1	1	Not present
First cached call, query 4	1	1	1	1	1

At $n = 2048$, the full prefill array has 4,194,304 entries per head, whereas the first cached call has 2,049. Eliminating complete intermediate arrays therefore offers much less score-storage saving for that cached call. It still reads the KV history. The full decoder also reads model weights, runs projections and feed-forward layers, and dispatches kernels. At batch size one, those costs can take a larger fraction of the step. The grid shapes suggest an explanation; measurements must establish the bottleneck.

First collect a kernel trace that separates attention, projections, and feed-forward work for each stage. Keep the model, input, batch size, precision, and device fixed while switching the attention implementation. Second, measure the attention kernels' HBM read and write traffic in separate profiling runs. A large drop in prefill writes with little change in decode reads would support the proposed explanation. If attention speeds up but occupies only a small part of the step, the trace explains the small overall improvement.

For the timing calculation, other work occupies $100 - 20 = 80$ ms. Halving attention leaves $20/2 = 10$ ms, so the new step takes **90 ms**. The overall speedup is $100/90 \approx 1.111$, or a 10% reduction in elapsed time. Even zero attention time would leave 80 ms. These are illustrative timings, not measured results for the NumPy example.

11. Plan a fair kernel comparison (optional)

This is a proposed measurement protocol, not a benchmark that has been run. Compare causal prefill attention on already projected Q, K, and V. The materializing path writes full scores and probabilities to device memory; the fused path carries tile data and running row statistics. In both paths, query i may read keys 0 through i, including its own key.

Use prompt lengths 512, 2048, and 4096, batch size two, eight heads, and head width 64. Choose one NVIDIA A100 GPU. Use FP16 inputs and outputs with FP32 softmax accumulation in both paths. Record intermediate formats too: a full score array stored in FP32 has four bytes per entry. Keep the tensor layout, head grouping, scale, and mask fixed; disable dropout. Prepare the same seeded, unit-scale projected arrays for both paths at each length. Record the software versions and the kernels that actually execute. An automatically fused reference would change the intended comparison.

Check correctness before timing. On a short prefix, compare each query and head with a float64 materializing reference that subtracts the row maximum before exponentiating. Verify the causal mask and perturb a future value to ensure earlier outputs remain unchanged. At each timed length, compare outputs on the same inputs. For these unit-scale examples, a proposed starting tolerance is $|x - y| \leq 2 \cdot 10^{-3} + 2 \cdot 10^{-3}|y|$ per output coordinate. Record maximum absolute differences and investigate failures before timing. This tolerance is an experimental choice for these inputs, not a universal bound for FP16 attention.

Time the attention operation. Inputs are already resident on the device when the timer starts; stop when the output is ready there. Include score calculation, masking, softmax, and the weighted sum, including intermediate writes, reads, tile processing, and any workspace allocation done per call. If constructing a mask is part of the call, include it. Leave input and output projections and token sampling outside this interval. A separate full-model prefill measurement can show the effect on time to first token.

Warm up both paths at each shape. Run 30 paired trials with their order alternating. Synchronize the device before starting and before stopping a wall-clock timer; this includes host work within the measured call. Report the median and interquartile range for each length. Do not mix asynchronous launch time with completed device execution time.

Measure memory in separate runs. Once inputs are resident, reset the peak allocated-memory counter and record the peak above that baseline. Use an allocation trace to separate temporary workspaces from the common output buffer at the same instants. Report reserved memory separately, since an allocator can retain unused blocks after tensors are released.

To test the data-movement explanation, collect hardware counters in separate warmed runs when the device supports them. Sum bytes read from and written to HBM by all kernels in one attention call. Compare at identical shapes. Peak workspace counts bytes stored at once; traffic counts bytes transferred over time. A reduction in both elapsed time and transferred bytes supports the explanation. Timing alone cannot distinguish that effect from better scheduling or fewer arithmetic operations.

Checks

All eleven exercises have complete solutions. The executed checks cover response times, aggregate throughput, exact byte counts, both tile orders, partial tiles, output invariance across tile shapes, empty masked rows, absolute query positions, and large-score stability. Numerical equality uses a declared tolerance; memory counts specify which arrays they include.

Next Steps

Change a tile shape or a query offset, predict the answer, then rerun. Treat the NumPy implementation as an inspectable correctness model. Its CPU timing does not measure a fused GPU kernel. The original FlashAttention paper and FlashAttention-2 paper explain the GPU schedules.